

Improving User QoE via Joint Trajectory and Resource Optimization in Multi-UAV Assisted MEC

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Abstract—As a promising network architecture, Mobile Edge Computing (MEC), has been proven that can effectively reduce the end-to-end latency and the energy consumption. The Unmanned Aerial Vehicle (UAV) assisted MEC network, where the UAV can provide the computation offloading services for the mobile users, can further alleviate the huge deployment cost of static edge servers. However, it remains unsolved how multiple cooperative flying UAVs serve the ground users, especially considering that these UAVs may share the same wireless channel and can communicate with the users while flying. In this paper, we first propose the Age of Task (AoT) metric to measure the quality of experience, and then formulate the joint optimization problem to minimize the worst AoT among all the users. Based on the block coordinate descent (BCD) method, this problem is transformed into three non-convex programming sub-problems (i.e., the UAV-user association sub-problem, the UAV trajectory planning sub-problem and the transmit power optimization sub-problem). Specifically, the successive convex approximation (SCA) technique is exploited iteratively to deal with the non-convexity in the UAV trajectory and transmit power optimization. Numerical results show that the proposed scheme outperforms the benchmark offloading schemes in terms of AoT.

Index Terms—Mobile edge computing, computation offloading, UAV trajectory.

I. INTRODUCTION

THERE are a large number of mobile applications (e.g., image analysis [1], target tracking [2], virtual reality [3] and online gaming [4]) requiring the low latency, the huge computational capacity and the low energy consumption in the real world. As a promising network architecture, mobile edge computing (MEC) [5], [6], which can provide the flexible and efficient computational services for the mobile users in geographically proximity areas, has been proved that can effectively decrease the end-to-end latency and the energy consumption [7], [8]. For example, partial or binary computation-intensive tasks can be offloaded to the nearby static edge servers, e.g., access points (APs) and cellular base stations (BSs) [9].

To alleviate the huge cost caused by deploying static edge servers densely, the Unmanned Aerial Vehicle (UAV) assisted MEC network has been introduced [10], [11], where a flying UAV is scheduled such that the computation offloading service can be provided to the mobile users. Specifically, UAVs can exploit the flexibility and ease of their deployment to enhance the wireless communication [12], assist the data collection [13], and support the monitoring coverage [14] in the MEC architectures. Moreover, UAVs will rarely be constrained by the spatiotemporal constraints, e.g., the road traffic congestion and the terrain obstruction, compared with the vehicular edge computing. Therefore, the UAV-assisted MEC has been becoming a promising technology, where UAV-user association, trajectory plan and power allocation are attracting research attention.

However, most of the existing works focus on the following situations: (1) a single UAV and multiple users scenario: only a single UAV [15] (with a jammer UAV considering secure communication [16]) serves multiple users; or (2) fly-hover-communication mode: the offloading services are conducted only when UAV hovers or hangs in the air [13], [17], [18]. A more general situation illustrated in Fig. 1, i.e., *How multiple flying UAVs cooperatively serve the ground users*, remains unsolved, especially considering that these UAVs *share the same wireless channel and communicate with the users while flying*. Intuitively, since more flying UAVs can provide more computation resources, it may further improve the quality of experience (QoE), e.g., the end-to-end latency. Thus it is non-trivial to jointly schedule multiple cooperative UAVs to offload the intensive computational tasks, especially when the shared channel may result in the co-channel interference. Moreover, the existing metric about the user's temporal experience (e.g., the

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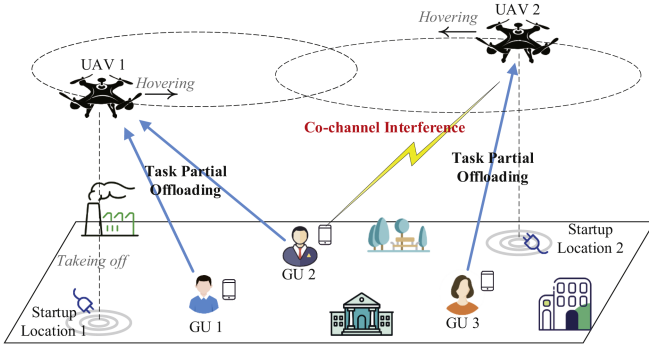


Fig. 1. Multiple flying UAVs serve the ground users (GUs) cooperatively.

task latency) can not clearly capture the fine-grained difference, especially in the partial task offloading scenario.

In this paper, we will formulate the joint optimization problem to minimize the worst corresponding experience among the users. Then the successive convex approximation technique and the block coordinate descent method are introduced to solve the UAV-user association sub-problem, the UAV trajectory planning sub-problem and the transmit power optimization sub-problem. The main contributions of this paper are summarized as follows.

- We formulate the user-utility maximum problem as a joint optimization problem with local/offloading task allocation, UAV trajectory and transmit power, in the multi-UAV multi-User UAV-assisted MEC scenario.
- We utilize the successive convex approximation technique and the block coordinate descent method to deal with the joint non-convexity, where the original problem is transformed into three convex sub-problems.
- We conduct extensive simulations to verify the proposed joint optimization scheme outperforms the benchmark schemes in terms of the worst user experience.

The remainder of this article is organized as follows. The related literature is discussed in Section II, and the mathematical model is presented in Section III. The proposed algorithm is described in Section IV. The computational results are given in Section V. This paper concludes in Section VI.

Notations: In this paper, scalars are denoted by italic letters, e.g., a . Vectors and matrices are denoted by boldface lower case letters and upper case letters, respectively, e.g., $\mathbf{a}$ and $\mathbf{A}$. For a vector $\mathbf{a}$, $\|\mathbf{a}\|$ denotes its l_2 norm. $\mathbf{a}^T$ (or $\mathbf{A}^T$) denotes the transpose of vector $\mathbf{a}$ (or matrix $\mathbf{A}$). For a time-relevant function $\mathbf{a}(t)$, $\dot{\mathbf{a}}(t)$ represents the first derivative of $\mathbf{a}(t)$ with respect to time t .

II. RELATED WORK

A. UAV-Assisted Communication

Based on the analysis of the communication throughput and the fixed-wing UAV energy consumption, [19] designed an energy-efficient UAV communication on the UAV trajectories, with the initial/final UAV location and velocity constraints. After formulating the rotary-wing UAV propulsion power model, Zeng et al. [20] proposed the energy minimization algorithm to optimize the UAV trajectory and the communication time allocation.

Moreover, [20] verified the advantage of the communicate-while-flying over the fly-hover-communicate design. To maximize the minimum average communication rate from UAVs to users, Wu et al. [21] investigated the multi-UAV wireless communication network and proposed the optimization algorithm based on the SCA technique.

B. Static Edge Server Enabled MEC

It is of great importance to allocate the resources and offload the tasks in the MEC network. In the multi-server MEC network, where each BS is equipped with an MEC server, [9] formulated the task offloading problem and maximized the offloading gain via decomposing the original problem into the resource allocation and task offloading sub-problem. A unified MEC-WPT system was designed in [22], where the AP not only broadcasts wireless power to charge multiple users but also executes the offloaded computation tasks from users. Considering the limited coverage of the static edge servers, Liu et al. [23] introduced a vehicle-mounted edge server and utilized the piecewise linear approximation to maximize the completed tasks via planning the vehicle path and allocating the resources. Based on the deep Q-network and the deep deterministic policy gradient, Jiang et al. [24] minimized the task computation latency for D2D-assisted MEC networks. Since the dependence among tasks may constrain the scheduling decisions, [25] exploited the direct acyclic graph (DAG) to represent this task dependence and designed the task ranking to minimize the average latency.

C. UAV-Assisted Mec

Due to the flexibility of the deployment and the improvement of communication channels, the UAV-assisted MEC architecture is introduced to provide the task offloading service for the ground users, especially in the extreme situations, where the conventional communication systems are destroyed by natural disasters or armed conflicts [26].

Some researchers focus on the fly-hover-communicate model of the UAV-assisted MEC network. In this situation, the UAVs will receive the task input data and conduct the offloaded tasks while hovering at the predetermined locations. To minimize the average time delay of the user without exceeding the maximum energy consumption, Liao et al. [17] dispatched the UAVs to the optimal hover positions and assisted the ground BS for task offloading. Based on the submodular optimization, the hovering locations are selected to maximize the total benefit in [18], which further generated a UAV flight trajectory. In order to optimize the system welfare and the energy consumption in the marine multi-UAV MEC, Dai et al. [27] proposed an efficient joint optimization method based on double auction and bisection/linear search to offload tasks from multiple UAVs to multiple ocean beacon stations.

The other researchers try to achieve the optimization objective in a more general scenario, where the UAVs can receive the data from the users while flying. [28] designed a UAV-enabled wireless powered MEC system, where a UAV can provide computational services for multiple users and transmit energy signals to charge them. With the fixed initial/final location, [11]

investigated the UAV energy minimization problem and the completion time minimization problem in the single UAV-enabled MEC system. To serve the ground users with an existing ground eavesdropper, [29] minimized the UAV energy consumption with the secure offloaded rate constraint. Besides a server UAV in the UAV-assisted MEC system, the second UAV was used by [16] as a jammer to suppress the ground eavesdroppers and improve the secure computing capacity. [30] adopted the Lyapunov optimization to achieve the long-term stability of the task queue, which will further simplify the sum energy minimization problem. In a single-vehicle and on-board UAV system, where the UAV needs to return to the vehicle, [6] facilitated the offloading computation services for the devices via optimizing the UAV trajectory, the computing resources and the communication scheduling. Deng et al. [31] eliminated the fractional structure based on Dinkelbach's method to maximize the computation efficiency. To improve the joint utility of the task priority and the energy consumption in the multi-UAV MEC system, deep reinforcement learning is exploited in [32] to efficiently design the UAV trajectory, make the offloading decisions and allocate the computation/communication resources. Considering the complexity of the multi-UAV scenario, the DAG task feature and the offloading during flying, [33] adopted the genetic algorithm to deal with the joint optimization problem, which effectively reduces the hover and flight energy consumption.

However, in the multi-UAV MEC, few works concern users' temporal experience and consider the following constraints simultaneously: (1) the UAV propulsion energy should be considered as a vital constraint for the long-term running of the multi-UAV assisted MEC; (2) the multiple wireless transmissions at the same time may induce the co-channel inference, which is a natural constraint in this multi-UAV scenario; (3) the UAVs during the flight can also serve the ground users cooperatively. Thus this paper will propose a new metric to describe the temporal performance in a fine-grained view and design the joint optimization scheme for it.

III. PROBLEM FORMULATION

In this section, we first introduce the system model, including the UAV mobility, the GU-UAV communication and the local/offloading computation. Thereafter, we formulate the optimization problem mathematically.

A. System Model

We consider a multi-UAV assisted MEC network, where M UAVs provide the task offloading services to K ground users (GUs) within a given time period of T , in Fig. 1. All the UAV are equipped with the co-located MEC server respectively, but share a same wireless communication channel.

The set of GUs is denoted as $\mathcal{K} = \{1, 2, \dots, K\}$. The coordinate of GU $k \in \mathcal{K}$ is $\mathbf{w}_k = [x_k, y_k]^T \in \mathbb{R}^{2 \times 1}$. Each GU holds a computation-intensive task, of which the task-input bits is L_k . The computation tasks can be arbitrarily partitioned into several parts, for local computing or for task offloading via UAVs. c_k (cycles/bit) is the number of CPU cycles that is required to complete one bit in GU k 's task.

The set of UAVs is denoted as $\mathcal{M} = \{1, 2, \dots, M\}$. It is assumed that all the UAVs fly at a fixed altitude H meters (m) after leaving their respective startup locations, as shown in Fig. 1. The time-varying horizontal coordinate of UAV $m \in \mathcal{M}$ at time t is denoted by $\mathbf{q}_m(t) = [x_m(t), y_m(t)]^T \in \mathbb{R}^{2 \times 1}$.

For ease of analysis, we discretize the overall period T into N small equal-size time slots (denoted by $\mathcal{N} = \{1, 2, \dots, N\}$) with the length $\delta = T/N$. Thus the trajectory of UAV m in T can be represented by $\{\mathbf{q}_m[n]\}_{n=1}^N$, where the location of UAV m in time slot n is $\mathbf{q}_m[n] = [x_m[n], y_m[n]]^T$. In addition, the UAVs need to return their respective predetermined startup locations [21] at the end of the period T , such that these UAVs can be recharged for the next period. For example, we assume that UAV m should return to its own startup location $\mathbf{q}_m^s$, i.e.,

$$\mathbf{q}_m[0] = \mathbf{q}_m[N+1] = \mathbf{q}_m^s, \quad \forall m \in \mathcal{M}. \quad (1)$$

Furthermore, as discussed in [21], the trajectories of UAVs should satisfy the maximum speed constraints and the collision avoidance constraints,

$$\|\dot{\mathbf{q}}_m(t)\| \leq v_{\max}, \quad 0 \leq t \leq T, \forall m \in \mathcal{M},$$

$$\|\mathbf{q}_m(t) - \mathbf{q}_j(t)\| \geq d_{\min}, \quad 0 \leq t \leq T, \forall j, m \in \mathcal{M}, j \neq m,$$

which imply that all the UAVs should never exceed the maximum flying velocity $v_{\max}$ (meters/second), and at least keep any other UAVs at distance ($\geq d_{\min}$ meters) for flying safety. Considering the discretized time slots with enough small size, we can approximately transform these two constraints into

$$\|\mathbf{q}_m[n+1] - \mathbf{q}_m[n]\| \leq \delta v_{\max}, \quad \forall n \in \mathcal{N}, \forall m \in \mathcal{M}, \quad (2a)$$

$$\|\mathbf{q}_m[n] - \mathbf{q}_j[n]\| \geq d_{\min}, \quad \forall n \in \mathcal{N}, \forall j, m \in \mathcal{M}, j \neq m. \quad (2b)$$

In this paper, we also adopt the TMEDA-based communication and offloading design [16], where each slot is divided into K smaller sub-slots for the candidate GUs. For a specific UAV, each sub-slot is allocated to each GU in order with the duration of $\delta\tau_{k,m}[n]$. Here $\tau_{k,m}[n]$ denotes the proportion of the time slot n in UAV m , which is assigned to GU k . Therefore, we have the association constraints, i.e.,

$$0 \leq \tau_{k,m}[n] \leq 1, \quad \forall k, m, n, \quad (3a)$$

$$\sum_{k=1}^K \tau_{k,m}[n] \leq 1, \quad \forall k, m, \quad (3b)$$

$$\sum_{m=1}^M \tau_{k,m}[n] \leq 1, \quad \forall k, n. \quad (3c)$$

Here (3b) and (3c) imply that the sum of these nonnegative allocated time can not exceed a single time slot for UAV m or GU k , respectively.

B. Communication Model

In the UAV-assisted MEC network, the task-input data in GUs can be transmitted to UAVs for further computation offloading. It is always assumed that the wireless communication channels between UAVs and GUs are dominated by the Line-of-Sight (LoS) links [21] due to the relatively high flying altitudes. Similar to [19], the Doppler effect caused by the UAV mobility is assumed to be compensated at the receivers in this paper. Thus

the channel power gain at UAV m from GU k in time slot n follows the free-space path loss model, i.e.,

$$h_{k,m}[n] = \frac{\rho_0}{H^2 + \|\mathbf{q}_m[n] - \mathbf{w}_k\|^2}, \quad \forall k, m, n, \quad (4)$$

where ρ_0 denotes the received channel power of UAV m when the distance between UAV m and GU k is 1m. The transmit power of GU k is denoted by $p_k[n]$, $0 \leq p_k[n] \leq P^{\max}$, where $P^{\max}$ is the peak transmit power.

Since all the channels between M UAVs and K GUs share a same wireless communication channel, it is necessary to consider the co-channel interference in the communication model. Therefore, the achievable data transmission rate (bits/second) from GU k to UAV m in time slot n can be defined as

$$R_{k,m}[n] = B \log_2 \left(1 + \frac{p_k[n]h_{k,m}[n]}{\sum_{v=1, v \neq k}^K p_v[n]h_{v,m}[n] + \sigma^2} \right), \quad \forall k, m, n, \quad (5)$$

where σ^2 represents the power of the additive white gaussian noise in the UAV. Note that both $h_{k,m}[n]$ and $h_{v,m}[n]$ in (5) can be calculated based on (4). $\sum_{v=1, v \neq k}^K p_v[n]h_{v,m}[n]$ denotes the received power at UAV m because of the co-channel interference from other GUs except GU k .

The GU in this paper can offload a part of its task [5] to the UAVs that equipped with computational and communication capacity. Considering that the size of the computation outcome is much smaller than that of the task-input data, we ignore the time for sending results back to GUs in this paper. If the allocated proportion in time slot n for the association between GU k and UAV m is $\tau_{k,m}[n]$, the corresponding data that is transmitted from GU k to UAV m can be expressed as

$$L_{k,m}^{Comm}[n] = \delta \tau_{k,m}[n] R_{k,m}[n], \quad \forall k, m, n. \quad (6)$$

Thus the total energy consumption caused by the data transmission in GU k can be computed as

$$E_k^{Comm} = \sum_{n=1}^N \sum_{m=1}^M \delta \tau_{k,m}[n] p_k[n], \quad \forall k. \quad (7)$$

C. Computation Model

In this subsection, we introduce the computation model of the ground user devices and the UAV-mounted edge servers, respectively.

1) *Local Computation*: Although with the limited computation capacity, the GUs can still conduct a part of their respective computation task locally. For the task in GU k , $l_k[n]$ denotes the proportion of the task-input data, which is allocated to the local computation during time slot n . The local computation ability of GU k can be represented by the maximum CPU-cycle frequency, i.e., $F_{GU,k}^{\max}$ (cycles/second). Thus the local computation proportion $l_k[n]$ should satisfy

$$c_k l_k[n] L_k \leq \delta F_{GU,k}^{\max}, \quad \forall k, n, \quad (8)$$

which implies that the allocated task-input data should be completed at least with the maximum local CPU frequency of GU k . If the dynamic frequency and voltage scaling (DVFS) [14] is adopted in the GUs, the practical CPU frequency can be

$\frac{c_k l_k[n] L_k}{\delta}$. Then the computation energy consumption of GU u for local computation can be calculated as

$$E_k^{Le} = \sum_{n=1}^N \frac{\kappa (c_k l_k[n] L_k)^3}{\delta^2}, \quad \forall k, \quad (9)$$

where κ is an energy constant related to GU k 's hardware architecture.

2) *Offloading Computation*: In order to improve the system performance or the user's QoE via the relatively higher computation capacity of the edge devices, the tasks of GUs can also be offloaded partially to UAVs (indeed UAV-mounted edge servers). The maximum CPU frequency is denoted as the $F_{UAV,m}^{\max}$ (cycles/second). Thus the allocated time $\tau_{k,m}[n]\delta$ and the transmission rate $R_{k,m}[n]$ for computation offloading (from GU k to UAV m in time slot n) should satisfy

$$c_k \tau_{k,m}[n] \delta R_{k,m}[n] \leq \tau_{k,m}[n] \delta F_{UAV,m}, \quad \forall k, m, n, \quad (10)$$

which implies that the received task-input data from GU k must be completed by UAV m during the allocated time. Since $\tau_{k,m}[n]\delta \geq 0$, (10) can be transformed into

$$c_k R_{k,m}[n] \leq F_{UAV,m}^{\max}, \quad \forall k, m, n. \quad (11)$$

Similarly, the total computation-related energy consumption in the UAV that caused by the task offloading is

$$\begin{aligned} E_m^{Oe} &= \sum_{n=1}^N \sum_{k=1}^K \kappa c_k \delta \tau_{k,m}[n] R_{k,m}[n] \left(\frac{c_k \delta \tau_{k,m}[n] R_{k,m}[n]}{\delta \tau_{k,m}[n]} \right)^2 \\ &= \sum_{n=1}^N \sum_{k=1}^K \kappa \delta \tau_{k,m}[n] (c_k R_{k,m}[n])^3, \quad \forall m, \end{aligned} \quad (12)$$

Moreover, the allocation time variables should also ensure that all the tasks can be completed in the period T , which is

$$\sum_{n=1}^N \left[\left(\sum_{m=1}^M \tau_{k,m}[n] \delta R_{k,m}[n] \right) + l_k[n] L_k \right] \geq L_k, \quad \forall k. \quad (13)$$

Based on N discrete time slots, GU k 's task latency depends on the time slot, in which the last part of this task will be processed. Assume that GUs' tasks are generated at the beginning of the whole period (i.e., at the beginning of the first slot). The task offloading latency from GU k to UAV m can be approximately calculated as

$$\begin{aligned} &\max n \delta, \\ &\text{if } \tau_{k,m}[n] \delta R_{k,m}[n] > 0. \end{aligned}$$

For example, $T = 100s$ and $N = 200$ (i.e., $\delta = 0.5s$). If UAV 1 only processes the last 1 Mb in the 90th slot, the offloading task latency is $90 \times 0.5s = 45s$. Since the task can be partially offloaded in M UAVs or be computed in GU k itself in these N slots, the latency of GU k 's task can be expressed as

$$\begin{aligned} &\max n \delta, \\ &\text{if } \left(\sum_{m=1}^M \tau_{k,m}[n] \delta R_{k,m}[n] \right) + l_k[n] L_k > 0. \end{aligned}$$

which is also the last time (i.e., slot n) that the task needs to be processed.

D. Energy Consumption Model

1) *Energy Consumption of Ground Users*: To maintain the long-term running of GUs, the total energy consumption of every GU can not exceed the predetermined limit $E_k^{\max}$. Because the energy consumption in a GU consists of the communication energy and the local computation energy, according to (7) and (9), we have

$$\sum_{n=1}^N \left[\frac{\kappa(c_k l_k[n] L_k)^3}{\delta^2} + \sum_{m=1}^M \delta \tau_{k,m}[n] p_k[n] \right] \leq E_{GU}^{\max}, \quad \forall k. \quad (14)$$

2) *Energy Consumption of UAVs*: The energy consumption in a UAV generally consists of the communication energy, the offloading computation energy and the propulsion energy. In this paper, the communication energy can be ignored because the task outcome bits is few. The limit of the computation offloading energy consumption in UAV m is $E_{OE}^{\max}$. Based on (12), we obtain the offloading energy constraint, i.e.,

$$\sum_{n=1}^N \sum_{k=1}^K \kappa \delta \tau_{k,m}[n] (c_k R_{k,m}[n])^3 \leq E_{OE}^{\max}, \quad \forall m. \quad (15)$$

The propulsion energy of UAVs is always the main part of the UAV energy consumption. Similar to [21], [30], we adopt the existing propulsion model for rotary-wing UAVs, i.e.,

$$E_m^{prop} = \delta \sum_{n=0}^N P_0 \left(1 + \frac{3 \|v_m[n]\|^2}{U_{tip}^2} \right) + \frac{1}{2} d_0 \rho A s \|v_m[n]\|^3 + P_i \left(\sqrt{1 + \frac{\|v_m[n]\|^4}{4v_0^4}} - \frac{\|v_m[n]\|^2}{2v_0^2} \right)^{\frac{1}{2}}, \quad (16)$$

where the first term and the third term are the blade profile power and the induced power in hovering status, respectively. The second term represents the parasite power. U_{tip} denotes the tip speed of the rotor blade. v_0 in (16) is known as the mean rotor induced velocity in hover. d_0 and s are the fuselage drag ratio and rotor solidity, respectively. ρ and A denote the air density and rotor disc area, respectively. Note that the first and second term increase quadratically and cubically with the UAV's velocity $\|v_m[n]\|$, respectively. Based on the discretized UAV trajectory $q_m[n]$, the velocity of UAV m , i.e., $\|v_m[n]\|$, can be approximately computed as

$$v_m[n] = \frac{\|q_m[n+1] - q_m[n]\|}{\delta}, \quad \forall m \in \mathcal{M}, 0 \leq n \leq N.$$

Thus we have the constraint about the UAV propulsion energy constraint, i.e.,

$$\delta \sum_{n=0}^N \left[P_0 \left(1 + \frac{3 \|v_m[n]\|^2}{U_{tip}^2} \right) + \frac{1}{2} d_0 \rho A s \|v_m[n]\|^3 + P_i \left(\sqrt{1 + \frac{\|v_m[n]\|^4}{4v_0^4}} - \frac{\|v_m[n]\|^2}{2v_0^2} \right)^{\frac{1}{2}} \right] \leq E_{prop}^{\max}, \quad \forall m. \quad (17)$$

E. Problem Formulation

To measure the information updating timeliness (i.e., freshness), Age of Information (AoI) has been introduced in the single-hop or multi-hop wireless networks recently [34]. For example, when the freshest information was generated at time t_0 and is received at time t , AoI is calculated as $t - t_0$. Considering that GUs wish their tasks (or the gathered information) to be processed as soon as possible, we construct a similar metric, i.e., Age of Task (AoT), for the task offloading scenario in this paper. For instance, when a bit of the task was generated in time t_0 , its corresponding AoT can be computed $t - t_0$ as if the given bit is completed at time t .

In order to simplify the problem formulation, we will approximately formulate AoT in the discretized period T . As discussed in Section III-A, the overall period has been divided into N time slots with equal time length δ . If we obtain how many bits are processed in time slot n , we can simplify the AoT analysis in the view of GU k 's task with size L_k .

First, for the local computation, $l_k[n] L_k$ bits are processed locally at GU k in slot n , where $l_k[n]$ is the optimization variable. Then we discuss the offloading computation from GU k to UAV m . Because GU k can be served by all the UAVs (including UAV m), $\tau_{k,m}[n] \delta$ can denote the allocated time for the association between UAV m and GU k . Here $\tau_{k,m}[n]$ is also the optimization variable. Based on their transmission rate $R_{k,m}[n]$ in slot n , $\tau_{k,m}[n] \delta R_{k,m}[n]$ bits in GU k can be transmitted to UAV m in slot n . Moreover, the inequality (10) ensures that these transmitted bits can be processed in time by UAV m in the duration $\tau_{k,m}[n] \delta$. Thus the total number of bits that can be processed in slot n is $(\sum_{m=1}^M \tau_{k,m}[n] \delta R_{k,m}[n]) + l_k[n] L_k$, which considers both the offloading computation (in all the UAVs) and the local computation (in GU k itself). From a bit perspective, AoT is calculated as the time a bit waits before being processed. For example, if 100 bits are processed during slot n , the AoT of each bit equals $n\delta$. The total AoT of these 100 bits is $100n\delta$. Thus, the AoT of GU k 's whole task can be formulated as

$$A_k = \sum_{n=1}^N n\delta \left[\left(\sum_{m=1}^M \tau_{k,m}[n] \delta R_{k,m}[n] \right) + l_k[n] L_k \right], \quad (18)$$

where the waiting time of a bit (that is processed in slot n) is approximated as $n\delta$ according to N discretized time slots. Note that all the GU tasks are generated at time $t = 0s$. In this paper, we try to minimize the maximum AoT among all the GUs, which can be formulated as follows.

$$\begin{aligned} (P1) \quad & \min_{\mathbf{A}, \mathbf{Q}, \mathbf{P}} \max_{k \in \mathcal{K}} A_k \\ \text{s.t.} \quad & /*\text{energy constraint}*/ \\ & (14), (15), (17) \\ & /*\text{association constraint}*/ \\ & (3a), (3b), (3c) \\ & /*\text{local computing constraint}*/ \\ & 0 \leq l_k[n] \leq 1, \quad \forall k, n, \end{aligned} \quad (19a)$$

$$\sum_{n=1}^N l_k[n] \leq 1, \quad \forall k, \quad (19b)$$

(8)

/*offloading computing constraint*/

$$0 \leq p_k[n] \leq P^{\max}, \quad \forall k, n$$

(11), (13)

/*mobility constraint*/

$$(1), (2a), (2b). \quad (19c)$$

The optimization variables in problem (P1) consist of the GU-UAV association and local computing variables $\mathbf{A} = \{\tau_{k,m}[n], l_k[n], \forall k, m, n\}$, the UAV trajectory variables $\mathbf{Q} = \{\mathbf{q}_m[n], \forall m, \forall n\}$, the GU transmit power $\mathbf{P} = \{p_k[n], \forall u, n\}$. It can be seen the original optimization problem (P1) is not a convex optimization problem and can not be solve directly via the classic convex optimization methods. For example, constraint (2b) is concave with respect to $\mathbf{q}_m[n], \forall m, n$. Because the transmission rate $R_{k,m}[n]$ is not convex with respect to $\mathbf{q}_m[n]$ and $p_k[n]$, the objective A_k is not convex. Since the variables $\mathbf{A}$, $\mathbf{Q}$ and $\mathbf{P}$ in problem (P1) are mutually independent, we can adopt the block coordinate descent (BCD) method [21], [35] to transform problem (P1) into three sub-problems. Due to the non-convexity of two sub-problems, they will further be transformed into a sequence of convex optimization problems via the successive convex approximation (SCA) [36] technique. Finally, the original problem (P1) will be solved based on the variable iteration and the standard convex optimization solvers, e.g., CVX [37].

Furthermore, we can analyze the lower and upper bound of $\max_{k \in \mathcal{K}} A_k$ in the multi-UAV assisted MEC network.

$$\text{Lemma 1: } \min_{k \in \mathcal{K}} \sum_{n=1}^{N_{tw}} n \delta C_{up} \leq \max_{k \in \mathcal{K}} A_k \leq \max_{k \in \mathcal{K}} \sum_{n=1}^{N_{up}} n \delta \frac{\delta F_{GU,k}^{\max}}{c_k}$$

Proof of Lemma 1 is presented in the Appendix, available online.

IV. ALGORITHM DESIGN

In this section, we propose an iterative algorithm for (P1) based on the BCD method, with which the original problem (P1) can be decomposed into three sub-problems, i.e., (P2), (P3) and (P4). Considering the non-convexity of problems (P3) and (P4), we transform them into two constrained convex problems (P3.1) and (P4.1) via the SCA technique, respectively. Fig. 2 shows the decomposition of the original problem and the transformation of the sub-problems. Finally, we present the optimization scheme, which updates the variables $\mathbf{A}$, $\mathbf{Q}$ and $\mathbf{P}$ to achieve the optimal decisions.

A. Sub-Problem 1: GU-UAV Association and Local Computation Optimization

When the GU transmit power $\mathbf{P}$ and the UAV trajectory $\mathbf{Q}$ are given, the original problem (P1) can be transformed into a sub-problem of optimizing $\mathbf{A}$. Here $R_{k,m}[n]$ can be

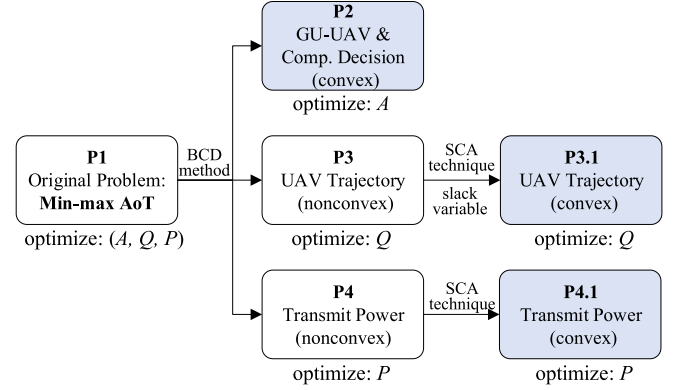


Fig. 2. Original problem decomposition and approximate transformation of sub-problems.

computed based on (5) with the given $p_k[n], \forall k, n$ and $\mathbf{q}_m[n], \forall m, n$. Note that we can find that the UAV mobility constraints (2a), (1) and (2b) are not relevant to this sub-problem about optimizing $\mathbf{A}$ ($\tau_{k,m}[n]$ and $l_k[n]$). Similarly, it is unnecessary to consider the UAV propulsion energy constraint (17) and the offloading frequency constraint (11) in this subsection. Thus the sub-problem about optimizing $\mathbf{A}$ can be

$$(P2) \quad \min_{\mathbf{A}} \max_{k \in \mathcal{K}} A_k$$

s.t. /*energy constraint*/

$$(14), (15)$$

/*association constraint*/

$$(3a), (3b), (3c)$$

/*local computing constraint*/

$$(19a), (19b), (8)$$

/*offloading computing constraint*/

$$(13).$$

Since $l_k[n] \geq 0$, the left-hand-side (LHS) of constraint (14) in problem (P2) is jointly convex with respect to $\tau_{k,m}[n]$ and $l_k[n]$. So its sub-level set, i.e., (14), constructs a convex set. The other constraints in problem (P2) are linear constraints. Moreover, the optimization objective related function A_k is also a linear function with respect to $\tau_{k,m}[n]$ and $l_k[n]$. Then the objective, min-max A_k , is also a convex function. Thus problem (P2) is a traditional convex optimization problem, which can be solve efficiently with the classic solvers/tools, e.g., SeDuMi, SDPT3, Mosek and CVX [37]. We can also infer that the inequality in the constraint (13) will be active at the optimal solution of $\mathbf{A}$, because processing more bits than L_k will increase the corresponding AoT A_k (18) in vain.

B. Sub-Problem 2: UAV Trajectory Optimization

According to (5), the UAV trajectory variable $\mathbf{Q}$ is relevant to the communication rate $R_{k,m}[n]$. However, whether the association constraints (3a) (3b) (3c) and the local computation constraints (19a) (19b) (8) are satisfied will not change due to the

variation of the optimization variable $\mathbf{Q}$. Similarly, the irrelevant constraints (14) and (13) are not necessary to be considered in this UAV trajectory optimization sub-problem. Therefore, for any given GU-UAV association $\mathbf{A}$ (i.e., $\tau_{k,m}[n]$, $l_k[n]$) and transmit power $\mathbf{P}$ (i.e., $p_k[n]$), the UAV trajectory optimization problem can be formulated as

$$\begin{aligned}
 (P3) \quad & \min_{\mathbf{Q}} \max_{k \in \mathcal{K}} A_k \\
 \text{s.t.} \quad & /*\text{energy constraint}*/ \\
 & (17), (15) \\
 & /*offloading computing constraint*/ \\
 & (11), (13) \\
 & /*mobility constraint*/ \\
 & (1), (2a), (2b).
 \end{aligned}$$

The LHS of constraint (2b) is a convex function with respect to $\mathbf{q}_m[n]$. Since the super-level set of a convex function is not convex in general, constraint (2b) is not a convex constraint. Due to the non-convexity and non-concavity of $R_{k,m}[n]$ with respect to $\mathbf{q}_m[n]$, the communication rate $R_{k,m}[n]$ in (5) and its relevant constraints (15) (11) (13) make problem (P3) challenging. Furthermore, the energy consumption constraint (17) in (P3) is not a convex constraint. Thus we exploit the SCA technique to transform the non-convex optimization problem into a series of convex optimization problems with the cost of the solution's accuracy. The main idea of the SCA technique [36] is as follows: (1) for any 'given value' of optimization variables, construct and solve a new convex optimization problem in the neighborhood area of these given values by replacing the constraints with their first-order Taylor expansions; (2) iteratively achieve the optimization result of the non-convex problem by replacing the 'given value' with the obtained solution at each iteration.

Thus the communication rate functions $R_{k,m}[n]$, the LHS of (2b) and (17) can be transformed into the approximate convex (indeed linear) constraints via the first-order Taylor expansion according to the SCA technique, respectively.

1) *Communication Rate Constraints With Co-Channel Interference*: The path-loss model in (4) is related to the UAV trajectory $\mathbf{q}_m[n]$, which has been discussed in [19], [38]. However, the co-channel interference, which results from the channel share of the multi-UAV multi-GU scenario in this paper, make this problem more complicated. Based on the channel gain (4), the transmission rate (5) can be rewritten as

$$\begin{aligned}
 R_{k,m}[n] &= B \log_2 \left(1 + \frac{p_k[n] h_{k,m}[n]}{\sum_{v=1, v \neq k}^K p_v[n] h_{v,m}[n] + \sigma^2} \right) \\
 &= B \log_2 \left(\frac{\sum_{v=1}^K \frac{p_v[n] \rho_0}{H^2 + \|\mathbf{q}_m[n] - \mathbf{w}_v\|^2} + \sigma^2}{\sum_{v=1, v \neq k}^K \frac{p_v[n] \rho_0}{H^2 + \|\mathbf{q}_m[n] - \mathbf{w}_v\|^2} + \sigma^2} \right) \\
 &= B \log_2 \left(\underbrace{\sum_{v=1}^K \frac{p_v[n] \rho_0}{H^2 + \|\mathbf{q}_m[n] - \mathbf{w}_v\|^2} + \sigma^2}_{\hat{R}_m[n]} \right)
 \end{aligned}$$

$$- B \log_2 \left(\underbrace{\sum_{v=1, v \neq k}^K \frac{p_v[n] \rho_0}{H^2 + \|\mathbf{q}_m[n] - \mathbf{w}_v\|^2} + \sigma^2}_{\tilde{R}_{k,m}[n]} \right), \quad (22)$$

where the first term and the second term are denoted by $\hat{R}_m[n]$ and $\tilde{R}_{k,m}[n]$, respectively.

Due to the non-convexity of $\hat{R}_m[n]$, we derive its first-order Taylor approximation with respect to the optimization variable $\mathbf{q}_m[n]$ at the given trajectory value $\mathbf{q}_m[n]^r$ as follows,

$$\begin{aligned}
 \hat{R}_m[n] &= B \log_2 \left(\sum_{v=1}^K \frac{p_v[n] \rho_0}{H^2 + \|\mathbf{q}_m[n] - \mathbf{w}_v\|^2} + \sigma^2 \right) \\
 &\geq B \log_2 \left(\sum_{v=1}^K \frac{p_v[n] \rho_0}{H^2 + \|\mathbf{q}_m[n]^r - \mathbf{w}_v\|^2} + \sigma^2 \right) \\
 &\quad - B \sum_{v=1}^K \left[\frac{\frac{p_v[n] \rho_0}{(H^2 + \|\mathbf{q}_m[n]^r - \mathbf{w}_v\|^2)^2} \log_2(e)}{\sum_{v=1}^K \frac{p_v[n] \rho_0}{H^2 + \|\mathbf{q}_m[n]^r - \mathbf{w}_v\|^2} + \sigma^2} \right. \\
 &\quad \left. 2(\mathbf{q}_m[n]^r - \mathbf{w}_v)^T (\mathbf{q}_m[n] - \mathbf{q}_m[n]^r) \right] \\
 &\triangleq \hat{R}_m^{lb}[n]. \quad (23)
 \end{aligned}$$

where the gradient of $\|\mathbf{q}_m[n] - \mathbf{w}_v\|^2$ (l_2 -norm's square) is

$$\frac{\partial \|\mathbf{q}_m[n] - \mathbf{w}_v\|^2}{\partial \mathbf{q}_m[n]} = 2(\mathbf{q}_m[n] - \mathbf{w}_v)^T. \quad (24)$$

Similarly, according to the first-order Taylor expansion with respect to the trajectory variable $\mathbf{q}_m[n]$, the second term in (22) can be approximately calculated as

$$\begin{aligned}
 \tilde{R}_{k,m}[n] &= B \log_2 \left(\sum_{v=1, v \neq k}^K \frac{p_v[n] \rho_0}{H^2 + \|\mathbf{q}_m[n] - \mathbf{w}_v\|^2} + \sigma^2 \right) \\
 &\geq B \log_2 \left(\sum_{v=1, v \neq k}^K \frac{p_v[n] \rho_0}{H^2 + \|\mathbf{q}_m[n]^r - \mathbf{w}_v\|^2} + \sigma^2 \right) \\
 &\quad - B \sum_{v=1, v \neq k}^K \left[\frac{\frac{p_v[n] \rho_0}{(H^2 + \|\mathbf{q}_m[n]^r - \mathbf{w}_v\|^2)^2} \log_2(e)}{\sum_{v=1, v \neq k}^K \frac{p_v[n] \rho_0}{H^2 + \|\mathbf{q}_m[n]^r - \mathbf{w}_v\|^2} + \sigma^2} \right. \\
 &\quad \left. 2(\mathbf{q}_m[n]^r - \mathbf{w}_v)^T (\mathbf{q}_m[n] - \mathbf{q}_m[n]^r) \right] \\
 &\triangleq \tilde{R}_{k,m}^{lb}[n]. \quad (25)
 \end{aligned}$$

Based on the approximations in (23) and (25), the transmission rate (22) can be approximately calculated as

$$\begin{aligned}
 R_{k,m}[n] &= \hat{R}_m[n] - \tilde{R}_{k,m}[n] \\
 &\approx \hat{R}_m^{lb}[n] - \tilde{R}_{k,m}^{lb}[n], \quad (26)
 \end{aligned}$$

which is a linear function with respect to the trajectory variable $\mathbf{q}_m[n]$ at the neighborhood of the given value $\mathbf{q}_m[n]^r$.

Then constraints (15), (11) and (13) in problem (P3) can be transformed into two linear (also convex) constraints, i.e.,

$$\sum_{n=1}^N \sum_{u=1}^U \kappa c_k^3 \delta \tau_{k,m}[n] (\hat{R}_m^{lb}[n] - \tilde{R}_{k,m}^{lb}[n])^3 \leq E_{Oe}^{\max}, \quad \forall m, \quad (27a)$$

$$c_k (\hat{R}_m^{lb}[n] - \tilde{R}_{k,m}^{lb}[n]) \leq F_m^{\max}, \quad \forall k, m, n, \quad (27b)$$

$$\sum_{n=1}^N \left[\left(\sum_{m=1}^M \tau_{k,m}[n] \delta (\hat{R}_m^{lb}[n] - \tilde{R}_{k,m}^{lb}[n]) \right) + l_k[n] L_k \right] \geq L_k, \quad \forall k. \quad (27c)$$

2) *Distance Constraint*: The SCA technique can be exploited again to deal with the non-convexity of constraint (2b) in problem (P3). Let $\mathbf{Q}^r = \{\mathbf{q}_m[n]^r, \forall m, n\}$ be the given UAV trajectory at the r th-iteration of the SCA optimization. Based on the first-order Taylor expansion with respect to $\mathbf{q}_m[n]$ and $\mathbf{q}_j[n]$, $\forall m \neq j$, we have

$$\begin{aligned} & \|\mathbf{q}_m[n] - \mathbf{q}_j[n]\|^2 \\ &= (\mathbf{q}_m[n] - \mathbf{q}_j[n])^T (\mathbf{q}_m[n] - \mathbf{q}_j[n]) \\ &\geq \|\mathbf{q}_m[n]^r - \mathbf{q}_j[n]^r\|^2 \\ &\quad + 2(\mathbf{q}_m[n]^r - \mathbf{q}_j[n]^r)^T (\mathbf{q}_m[n] - \mathbf{q}_m[n]^r) \\ &\quad + 2(\mathbf{q}_m[n] - \mathbf{q}_j[n]^r)^T (-1)(\mathbf{q}_j[n] - \mathbf{q}_j[n]^r) \\ &\geq -\|\mathbf{q}_m[n]^r - \mathbf{q}_j[n]^r\|^2 \\ &\quad + 2(\mathbf{q}_m[n]^r - \mathbf{q}_j[n]^r)^T (\mathbf{q}_m[n] - \mathbf{q}_j[n]), \end{aligned} \quad (28)$$

where $\mathbf{q}_m[n]^r$ and $\mathbf{q}_j[n]^r$ are the given value. Thus the constraint (2b) can be approximately transformed into a linear constraint at the neighborhood of the given values, i.e.,

$$\begin{aligned} d_{\min}^2 &\leq -\|\mathbf{q}_m[n]^r - \mathbf{q}_j[n]^r\|^2 \\ &\quad + 2(\mathbf{q}_m[n]^r - \mathbf{q}_j[n]^r)^T (\mathbf{q}_m[n] - \mathbf{q}_j[n]), \quad \forall m \neq j, n. \end{aligned} \quad (29)$$

3) *Propulsion Energy Constraint*: Since that the third term in (16) is non-convex, constraint (17) in problem (P3) is not a convex constraint. Based on (17), the UAV velocity $v_m[n] \geq 0$ is always satisfied. According to [20], we also introduce a slack variable $y_m[n]$, i.e.,

$$y_m[n] = \left(\sqrt{1 + \frac{\|\mathbf{v}_m[n]\|^4}{4v_0^4}} - \frac{\|\mathbf{v}_m[n]\|^2}{2v_0^2} \right)^{\frac{1}{2}}, \quad \forall m, n, \quad (30)$$

where $y_m[n] \geq 0$. Obviously, (30) is equivalent to

$$y_m[n]^2 + \frac{\|\mathbf{v}_m[n]\|^2}{v_0^2} = \frac{1}{y_m[n]^2}. \quad (31)$$

Here the UAV propulsion energy consumption constraint (17) in problem (P3) can be transformed into

$$\begin{aligned} E_{prop}^{\max} &\geq \delta \sum_{n=0}^N \left[P_0 \left(1 + \frac{3\|\mathbf{v}_m[n]\|^2}{U_{tip}^2} \right) + \frac{1}{2} d_0 \rho A s \|\mathbf{v}_m[n]\|^3 \right. \\ &\quad \left. + P_i y_m \right], \quad \forall m, \end{aligned} \quad (32a)$$

$$y_m[n]^2 + \frac{\|\mathbf{v}_m[n]\|^2}{v_0^2} \geq \frac{1}{y_m[n]^2}, \quad \forall m, n, \quad (32b)$$

$$y_m[n] \geq 0, \quad \forall m, n, \quad (32c)$$

$$v_m[n] = \frac{\|\mathbf{q}_m[n+1] - \mathbf{q}_m[n]\|}{\delta}, \quad \forall m, n. \quad (32d)$$

If the strict inequality in constraint (32b) occurs, we can reduce $y_m[n]$ again and make (32b) satisfied with the strict equality. Meanwhile, the right-hand-side (RHS) of constraint (32a), which will also decrease with the decreasing $y_m[n]$, is still satisfied. Therefore, the original constraint (17) is satisfied if the new constraints in (32) are satisfied. However, (32b) is not a convex constraint (or set). Thus by adopting the first-order Taylor expansion for the LHS of (32b) with respect to $y_m[n]$ and $\mathbf{q}_m[n]$, we have the following inequality

$$\begin{aligned} & y_m[n]^2 + \frac{\|\mathbf{v}_m[n]\|^2}{v_0^2} \\ &= y_m[n]^2 + \frac{\|\mathbf{q}_m[n+1] - \mathbf{q}_m[n]\|^2}{\delta^2 v_0^2} \\ &\geq (y_m[n]^r)^2 + 2y_m[n]^r (y_m[n] - y_m[n]^r) \\ &\quad - \frac{\|\mathbf{q}_m[n+1]^r - \mathbf{q}_m[n]^r\|^2}{\delta^2 v_0^2} \\ &\quad + \frac{2}{\delta^2 v_0^2} (\mathbf{q}_m[n+1]^r - \mathbf{q}_m[n]^r)^T (\mathbf{q}_m[n+1] - \mathbf{q}_m[n]) \\ &\triangleq Y_m^{lb}[n]. \end{aligned} \quad (33)$$

Based on the lower bound of LHS of (32b), i.e., $Y_m^{lb}[n]$, the non-convex constraint (32b) can be replaced with

$$Y_m^{lb}[n] \geq \frac{1}{y_m[n]^2}. \quad (34)$$

Furthermore, the objective related function A_k in problem (P3), which is relevant to $R_{k,m}[n]$ according to (18), is not a convex function. Based on the (26), we replace $R_{k,m}[n]$ in (18) with $\hat{R}_m^{lb}[n] - \tilde{R}_{k,m}^{lb}[n]$ and then obtain the approximate convex objective-related function, i.e.,

$$\tilde{A}_k \approx \sum_{n=1}^N n \left[\left(\sum_{m=1}^M \tau_{k,m}[n] \delta (\hat{R}_m^{lb}[n] - \tilde{R}_{k,m}^{lb}[n]) \right) + l_k[n] L_k \right]. \quad (35)$$

As discussed in [37], the min-max form of convex functions is also a convex function.

Therefore, by replacing the non-convex constraints in problem (P3) with their approximate functions/constraints, the trajectory optimization problem (P3) can be transformed into

$$\begin{aligned} (P3.1) \quad & \min_{\mathbf{Q}} \max_{k \in \mathcal{K}} \tilde{A}_k \\ \text{s.t.} \quad & \text{/*energy constraint*/} \\ & E_{prop}^{\max} \geq \delta \sum_{n=0}^N \left[P_0 \left(1 + \frac{3\|\mathbf{q}_m[n+1] - \mathbf{q}_m[n]\|^2}{\delta^2 U_{tip}^2} \right) \right. \end{aligned}$$

$$+ \frac{d_0 \rho A s}{2\delta^3} \|\mathbf{q}_m[n+1] - \mathbf{q}_m[n]\|^3 + P_i y_m[n], \quad \forall m \quad (36a)$$

$$Y_m^{lb}[n] \geq \frac{1}{y_m[n]^2}, \quad \forall m, n \quad (36b)$$

$$y_m[n] \geq 0, \quad \forall m, n \quad (36c)$$

$$\sum_{n=1}^N \sum_{u=1}^U \kappa c_k^3 \delta \tau_{k,m}[n] (\hat{R}_m^{lb}[n] - \tilde{R}_{k,m}^{lb}[n])^3 \leq E_{Oe}^{\max}, \quad \forall m, \quad (36d)$$

/*offloading computing constraint*/

$$c_k(\hat{R}_m^{lb}[n] - \tilde{R}_{k,m}^{lb}[n]) \leq F_m^{\max}, \quad \forall k, m, n, \quad (36e)$$

$$\sum_{n=1}^N \left[\left(\sum_{m=1}^M \tau_{k,m}[n] \delta (\hat{R}_m^{lb}[n] - \tilde{R}_{k,m}^{lb}[n]) \right) + l_k[n] L_k \right] \geq L_k, \quad \forall k, \quad (36f)$$

/*mobility constraint*/

$$\|\mathbf{q}_m[n+1] - \mathbf{q}_m[n]\| \leq \delta v_{\max}, \quad \forall n, m, \quad (36g)$$

$$\mathbf{q}_m[0] = \mathbf{q}_m[N] = \mathbf{q}_m^s, \quad \forall m, \quad (36h)$$

$$d_{\min}^2 \leq -\|\mathbf{q}_m[n]^r - \mathbf{q}_j[n]^r\|^2 + 2(\mathbf{q}_m[n]^r - \mathbf{q}_j[n]^r)^T (\mathbf{q}_m[n] - \mathbf{q}_j[n]), \quad \forall m \neq j, n, \quad (36i)$$

/* neighborhood area constraint*/

$$\|\mathbf{q}_m[n] - \mathbf{q}_m[n]^r\|^2 \leq d_q^2, \quad \forall m, n, \quad (36j)$$

where constraint (36j) ensures that the new solution can only be obtained from the neighborhood of the given value $\mathbf{Q}^r$ (i.e., $\mathbf{q}_m[n]^r$) based on the maximum distance d_q^2 . The non-convex constraint (17) has been replaced with three convex constraints (36a), (36b) and (36c). Similarly, (15) has been replaced with (36d). (2b) is replaced with (36i).

In addition, we should provide the initial trajectory $\mathbf{Q}^0$ as the 'given value' in the first iteration of problem (P3.1). After conducting the first trajectory optimization, we obtain the optimization result $\mathbf{q}^*$, which will be the 'given value' of the second iteration. In this way, we try to achieve the optimal UAV trajectory for problem (P3).

Similar to problem (P2) in Section IV-A, problem (P3.1) in this subsection, i.e., the approximation transformation form of problem (P3), is also a convex optimization problem with several convex constraints. Thus we can also exploit the classic convex solver or tool to obtain its optimal solution, i.e., the optimal trajectories of the m UAVs, to jointly minimize the maximum AoT of K GUs, if the other two sets of optimization variables ($\mathbf{A}$ and $\mathbf{P}$) are given.

C. Sub-Problem 3: Transmit Power Optimization

Assume that $\mathbf{A}$ (including $\tau_{k,m}[n]$ and $l_k[n]$), $\mathbf{Q}$ ($\mathbf{q}_m[n]$) are known. The overall original (P1) for optimizing the transmit

power $\mathbf{P}$ (i.e., $p_k[n]$) can be formulated as

$$(P4) \quad \min_{\mathbf{P}} \max_{k \in K} A_k$$

s.t. /*energy constraint*/

$$(14), (15)$$

/*offloading computing constraint*/

$$(19c), (11), (13).$$

However, the transmit power variable $p_k[n]$ is relevant to the transmission rate $R_{k,m}[n]$ according to (5). Both $\hat{R}_m[n]$ and $\tilde{R}_{k,m}[n]$ in the communicate rate function (22) are the concave function with respect to $p_k[n]$. Thus constraint (15) (11) and (13) in problem (P4) is not convex, where $R_{k,m}[n] = \hat{R}_m[n] - \tilde{R}_{k,m}[n]$. Based on the analysis of the transmission rate function (22) in Section IV-B, we can derive the first-order Taylor expansion of $\hat{R}_m[n]$ and $\tilde{R}_{k,m}[n]$ with respect to $p_k[n]$, respectively. For $\hat{R}_m[n]$, we have

$$\begin{aligned} \hat{R}_m[n] &= B \log_2 \left(\sum_{v=1}^K \frac{p_v[n] \rho_0}{H^2 + \|\mathbf{q}_m[n] - \mathbf{w}_v\|^2} + \sigma^2 \right) \\ &\leq B \log_2 \left(\sum_{v=1}^K \frac{p_v[n]^r \rho_0}{H^2 + \|\mathbf{q}_m[n] - \mathbf{w}_v\|^2} + \sigma^2 \right) \\ &\quad + B \sum_{v=1}^K \left[\frac{\frac{\rho_0}{H^2 + \|\mathbf{q}_m[n] - \mathbf{w}_v\|^2} \log_2(e)}{\sum_{v=1}^K \frac{p_v[n]^r \rho_0}{H^2 + \|\mathbf{q}_m[n] - \mathbf{w}_v\|^2} + \sigma^2} \right. \\ &\quad \left. (p_v[n] - p_v[n]^r) \right] \\ &\triangleq \hat{R}_m^{ub}[n], \end{aligned} \quad (37)$$

where $\hat{R}_m^{ub}[n]$, as a approximate and upper bound of $\hat{R}_m[n]$, is a linear (also convex) function with respect to the transmit power variable $p_k[n]$. According to (22), $\tilde{R}_{k,m}[n]$ is a concave function with respect to $p_k[n]$. Thus for $p_k[n]$, the approximate rate $\hat{R}_m^{ub}[n] - \tilde{R}_{k,m}[n]$ is a convex function. Constraints (15) and (11) can be approximately transformed into

$$\sum_{n=1}^N \sum_{u=1}^U \kappa c_k^3 \delta \tau_{k,m}[n] (\hat{R}_m^{ub}[n] - \tilde{R}_{k,m}[n])^3 \leq E_{Oe}^{\max}, \quad \forall m, \quad (38a)$$

$$c_k(\hat{R}_m^{ub}[n] - \tilde{R}_{k,m}[n]) \leq F_{UAV,m}^{\max}, \quad \forall k, m, n. \quad (38b)$$

Because the LHS of (38a) is the convex function, constraint (38a), as a sub-level set of a convex function [37], becomes a convex constraint. Similarly, we can deduce constraint (38b) is also a convex constraint.

For $\tilde{R}_{k,m}[n]$ in (22), we can deduce that

$$\begin{aligned} \tilde{R}_{k,m}[n] &= B \log_2 \left(\sum_{v=1, v \neq k}^K \frac{p_v[n] \rho_0}{H^2 + \|\mathbf{q}_m[n] - \mathbf{w}_v\|^2} + \sigma^2 \right) \\ &\leq B \log_2 \left(\sum_{v=1, v \neq k}^K \frac{p_v[n]^r \rho_0}{H^2 + \|\mathbf{q}_m[n] - \mathbf{w}_v\|^2} + \sigma^2 \right) \end{aligned}$$

$$\begin{aligned}
& + B \sum_{v=1, v \neq u}^U \left[\frac{\frac{\rho_0}{H^2 + \|\mathbf{q}_m[n] - \mathbf{w}_v\|^2} \log_2(e)}{\sum_{v=1, v \neq k}^U \frac{p_v[n]^r \rho_0}{H^2 + \|\mathbf{q}_m[n] - \mathbf{w}_v\|^2} + \sigma^2} \right. \\
& \quad \left. (p_v[n] - p_v[n]^r) \right] \\
& \triangleq \tilde{R}_{k,m}^{ub}[n], \tag{39}
\end{aligned}$$

where $\tilde{R}_{k,m}^{ub}[n]$ is also a linear function with respect to $p_k[n]$. Thus constraint (13) can be transformed into

$$\begin{aligned}
\sum_{n=1}^N \left[\left(\sum_{m=1}^M \tau_{k,m}[n] \delta(\hat{R}_m[n] - \tilde{R}_{k,m}^{ub}[n]) \right) + l_k[n] L_k \right] \\
\geq L_k, \quad \forall k. \tag{40}
\end{aligned}$$

Furthermore, based on (18), the min-max objective A_k in problem (P4) should also be approximately transformed into a convex function with respect to $p_k[n]$, i.e.,

$$\begin{aligned}
\check{A}_k &= \sum_{n=1}^N n \left[\left(\sum_{m=1}^M \tau_{k,m}[n] \delta R_{k,m}[n] \right) + l_k[n] L_k \right] \\
&\approx \sum_{n=1}^N n \left[\left(\sum_{m=1}^M \tau_{k,m}[n] \delta(\hat{R}_m^{ub}[n] - \tilde{R}_{k,m}[n]) \right) + l_k[n] L_k \right]. \tag{41}
\end{aligned}$$

Therefore, the transmit power optimization problem (P4) can be transformed into a convex problem in the neighborhood of the given value $p_k[n]^r$, which is

$$(P4.1) \quad \min_{\mathbf{P}} \max_{k \in \mathcal{K}} \check{A}_k$$

s.t. /*energy constraint*/

$$\sum_{n=1}^N \left[\frac{\kappa(c_k l_k[n] L_k)^3}{\delta^2} + \sum_{m=1}^M \tau_{k,m}[n] \delta p_k[n] \right] \leq E_{GU}^{\max}, \quad \forall k, \tag{42a}$$

$$\sum_{n=1}^N \sum_{u=1}^U \kappa c_k^3 \delta \tau_{k,m}[n] (\hat{R}_m^{ub}[n] - \tilde{R}_{k,m}[n])^3 \leq E_{Oe}^{\max}, \quad \forall m, \tag{42b}$$

/*offloading computing constraint*/

$$0 \leq p_k[n] \leq P^{\max}, \quad \forall k, n \tag{42c}$$

$$c_k(\hat{R}_m^{ub}[n] - \tilde{R}_{k,m}[n]) \leq F_{UAV,m}^{\max}, \quad \forall k, m, n, \tag{42d}$$

$$\begin{aligned}
\sum_{n=1}^N \left[\left(\sum_{m=1}^M \tau_{k,m}[n] \delta(\hat{R}_m^{ub}[n] - \tilde{R}_{k,m}^{ub}[n]) \right) \right. \\
\left. + l_k[n] L_k \right] \geq L_k, \quad \forall k, \tag{42e}
\end{aligned}$$

$$\|p_k[n] - p_k[n]^r\|^2 \leq d_p^2. \tag{42f}$$

Considering that the approximation of the Taylor expansion performs well in the neighborhood of the given value, we

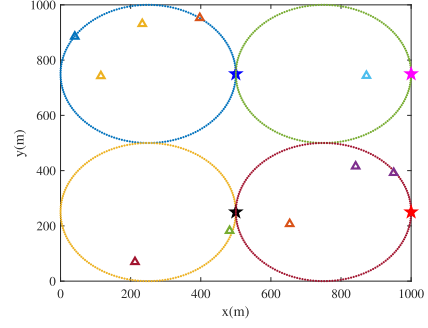


Fig. 3. Initial Trajectories of $M = 4$ UAVs in the example of $K = 10$ GUs.

introduce the neighborhood area constraint (42f) for the iterative conduction of problem (P4.1).

Based on the BCD method and the SCA technique, the overall optimization problem (P1) can be divided and transformed into three convex sub-problems, i.e., (P2), (P3.1) and (P4.1). By conducting these optimization sub-problems, we can propose the overall optimization algorithm, which is shown in Alg. 1. In each iteration of Alg. 1, the UAV trajectory $\mathbf{Q}$ (from Line 4 to Line 11), the transmit power $\mathbf{P}$ (from Line 13 to Line 19) and the GU-UAV association $\mathbf{A}$ (Line 21) are optimized successively. The maximum number of iterations is I , as shown in Line 2. According to the SCA technique, in order to obtain the optimal trajectory $\mathbf{Q}$, we conduct the approximate convex optimization problem (P3.1) iteratively until the corresponding objective value V_Q converges, i.e., V_Q stops decreasing or the decrement is less than threshold ϵ (Line 10). Similarly, problem (P4.1) is conducted iteratively based on the SCA technique.

Although Algorithm 1 depends on solving the above three convex sub-problems, designing an optimization algorithm for the standard convex problem is not the main focus of this paper. In our experiments, we adopt the Mosek optimizer to solve the three convex sub-problems. The implementation of the whole algorithm¹ is shown in our GitHub repository². However, in order to introduce the proposed method completely, we briefly add the details about solving the convex optimization problem in the Appendix, available online of this paper based on the method in [39], because the adopted solver Mosek is not an open-source solver.

V. PERFORMANCE EVALUATION

In this section, we evaluate the proposed algorithm via experiments using Matlab². We consider a 1000×1000 m area, where K GUs are randomly distributed. Each GU holds a task with the task-input data size of L_k bits. The UAV's flight altitude H is set as 100 m. The maximum permissible speed of UAVs, i.e., $v_{\max}$, is assumed as 30 m/s. The secure distance between any two UAVs, i.e., $d_{\min}$, is 100 m. The communication bandwidth is set as $B = 1$ MHz. According to [16] and [30], the noise power

¹https://github.com/gaoyang23nj/mUAV_MEC/blob/main/SolveP1bcdsca3subs.m

²https://github.com/gaoyang23nj/mUAV_MEC

Algorithm 1: Joint Optimization of GU-UAV Association, UAV Trajectory and GU Transmit Power and.

```

1: Initialize the feasible solution  $\mathbf{A}$ ,  $\mathbf{Q}$  and  $\mathbf{P}$ 
2: for  $i = 1 : I$  do
3:   /* Optimize  $\mathbf{Q}$  */
4:    $r \leftarrow 0$ ,  $\mathbf{Q}^0 \leftarrow \mathbf{Q}$ 
5:   repeat
6:     Calculate  $y_m[n]^r$  with  $\mathbf{Q}^r$  according to (30)
7:     Solve problem (P3.1) with given
        $\{\mathbf{A}, \mathbf{Q}^r, \mathbf{P}, y_m[n]^r\}$ , denote the solution as  $\mathbf{Q}^*$  and
       denote the objective value as  $V_Q$ .
8:      $\mathbf{Q}^{r+1} \leftarrow \mathbf{Q}^*$ 
9:      $r \leftarrow r + 1$ 
10:  until  $V_Q$  stop decreasing or the decrease of  $V_Q$  is
       below the threshold  $\epsilon > 0$ .
11:   $\mathbf{Q} \leftarrow \mathbf{Q}^{r+1}$ 
12:  /* Optimize  $\mathbf{P}$  */
13:   $r \leftarrow 0$ ,  $\mathbf{P}^0 \leftarrow \mathbf{P}$ 
14:  repeat
15:    Solve problem (P4.1) with given  $\{\mathbf{A}, \mathbf{Q}, \mathbf{P}^r\}$ ,
       denote the solution as  $\mathbf{P}^*$  and denote the objective
       value as  $V_P$ .
16:     $\mathbf{P}^{r+1} \leftarrow \mathbf{P}^*$ 
17:     $r \leftarrow r + 1$ 
18:  until  $V_P$  stop decreasing or the decrease of  $V_P$  is
       below the threshold  $\epsilon > 0$ .
19:   $\mathbf{P} \leftarrow \mathbf{P}^{r+1}$ 
20:  /* Optimize  $\mathbf{A}$  */
21:  Solve problem (P2) with given  $\{\mathbf{Q}, \mathbf{P}\}$ , denote the
       optimal solution as  $\mathbf{A}$  and denote the objective value
       as  $V$ .
22: end for

```

σ^2 and the reference power gain (at a distance of 1 m) ρ_0 are set as -110 dBm and -30 dBm, respectively. The UAV offloading computation energy constraint $E_{Oe}^{\max}$ is also 2,000 J. The GU energy consumption constraint $E_{GU}^{\max}$ is 200J. Unless otherwise specified, the UAV propulsion energy constraint $E_{prop}^{\max}$ is 50,000 J. K is set as 10. The rest of the parameters are presented in Table I.

There are 4 UAVs (i.e., $M = 4$), whose horizontal startup locations, i.e., $\mathbf{q}_m^s$, are (500, 250), (500, 750), (1000, 250) and (1000, 750) m, respectively. Before conducting the proposed algorithm (Alg. 1), we set (1) the initial transmit power as $P^{\max}$; (2) the initial GU-UAV association, i.e., $\tau_{k,m}[n] = \frac{1}{M}$; (3) the initial UAV trajectories, which are shown in Fig. 3. Here the startup locations of UAVs are denoted by symbol ‘★’. The location of GU is denoted by a ‘△’. The initial trajectory of a UAV is indeed an anticlockwise circle with a radius of 250 m, which starts and ends at its own startup location for further charging or daily maintenance.

A. Convergence of the Proposed Algorithm

Fig. 4 illustrates the convergence performance of the proposed algorithm (Alg. 1). We conduct Alg. 1 100 times with $M = 4$

TABLE I
SIMULATION PARAMETERS

Parameter	Description	Value
U_{tip}	Tip speed of the rotor blade (m/s)	120
P_0	Parameter in the First term of (16)	79.8573
P_i	Parameter in the Second term of (16)	88.6269
d_0	Fuselage drag ratio	0.6
ρ	Air density (kg/m^3)	1.225
A	Rotor disc area (m^2)	0.503
s	Rotor solidity	0.05
v_0	Mean rotor induced velocity (m/s)	4.03
P^{max}	Maximum transmit power of GUs (W)	0.1
H	Flying altitude of UAVs (m)	100
B	Communication bandwidth (MHz)	1
σ^2	Noise power (dBm)	-100
ρ_0	Reference power gain (dBm)	-30
$F_{GU,k}^{max}$	Maximum local CPU frequency in GU (GHz)	1
$F_{UAV,m}^{max}$	Maximum CPU frequency in UAV (GHz)	10
κ	Energy coefficient related to CPU	10^{-27}
c_k	Required cycles per bit computation (cycles/bit)	10^3
T	Overall period (s)	100
N	Number of time slots	200

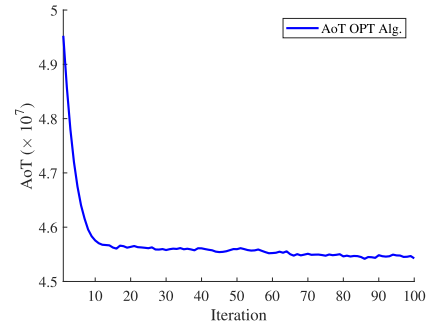


Fig. 4. Convergence of the optimization objective (i.e., min-max AoT).

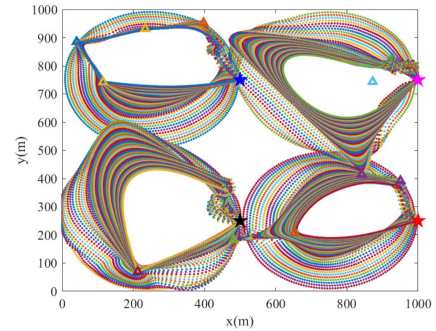


Fig. 5. Convergence of the UAVs trajectories when $K = 10$.

UAVs, $K = 10$ randomly distributed GUs and $L_k = 10$ Mb. According to (18), the AoTs of all the GUs at every iteration of Alg. 1 can be calculated. We can find that the achieved maximum AoT among the GUs, i.e., the min-max AoT, decreases quickly with iteration increasing from 1 to 15. Thereafter, the objective value of the proposed algorithm can always converge to around 4.5×10^7 within 50 iterations.

When the locations of GUs and the other parameters (including communication, computation and UAV flying parameters) are given, the proposed algorithm can update the UAV trajectory for each UAV when the iteration increases, which is shown in Fig. 5. For the UAV trajectory in every time slot,

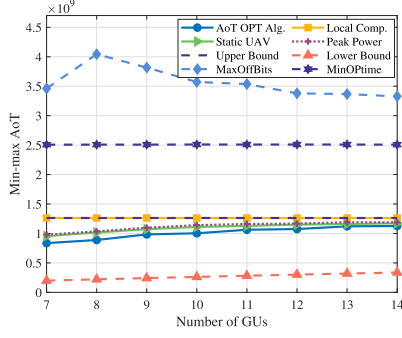


Fig. 6. Maximum AoT with varying $K \in [7, 14]$ when $L_k = 50$ Mb.

we sample one horizontal coordinate marked by one dot, i.e., $q_m[n]$, $1 \leq n \leq N$. Due to the neighborhood area constraint (36j), the trajectories in two subsequent iterations seem to be very close to each other, which results in the dense dots in Fig. 5. However, the neighbor area constraint will also be updated in the next iteration according to the SCA technique. Therefore, the proposed algorithm can successively optimize the UAV trajectory in terms of the min-max AoT.

B. Min-Max AoT Versus Number of UAVs K

Fig. 6 presents the min-max AoT of the algorithms with the varying number of GUs K , when the size of task-input bits L_k is set as 50 Mb. Here the min-max AoT is obtained based on the random distributed GU locations. We also evaluate five benchmark schemes: (1) Local Comp., with which the GU *completes its own task locally* and does not obtain the UAV-assisted computation service; (2) Static UAV, where all the UAVs *stay at their startup locations* $q_m[0]$ and provide the task offloading services with the optimized GU-UAV association and transmit power; (3) Peak Power, in which the GUs transmit tasks to UAVs but *with the fixed transmit power* $P_{\max}$; (4) MaxOffBits [40], which maximizes the amount of the bits that offloaded to UAVs by adjusting UAV trajectories, the task allocation and the transmit power; (5) MinOptime [41], which minimizes the total time cost (including the communication time and the computation time) via optimizing the offloading decisions, the UAV trajectories and the GU transmit power.

As the number of GUs K increases, the maximum AoT of all the GUs provided by Static UAV, Peak Power and the proposed algorithm (i.e., AoT OPT Alg.) deteriorates from less than 1×10^9 ($K = 5$) to more than 1.1×10^9 ($K = 14$). This is because the average available communication & computation resources for each GU continuously decrease when more GUs participate in the UAV-assisted MEC and share the offloading service. Moreover, the proposed algorithm is always superior to other schemes in terms of AoT due to its joint optimization of UAV trajectories and resources. Based on Lemma 1, the ‘Upper Bound’ and ‘Lower Bound’ of AoT can also be calculated and drawn in Fig. 6. We can also find that the min-max AoT of the proposed scheme always falls between the ‘Upper Bound’ and ‘Lower Bound’ which conforms to Lemma 1. Note that neither MaxOffBits nor MinOptime provides a relatively small

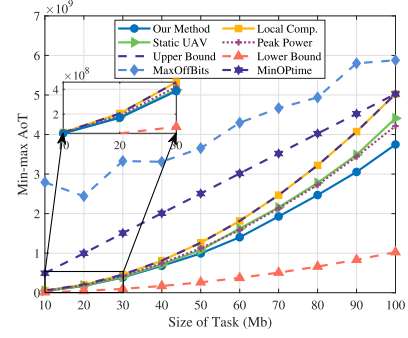


Fig. 7. Maximum AoT with varying $L_k \in [10, 100]$ Mb when $K = 10$ GUs.

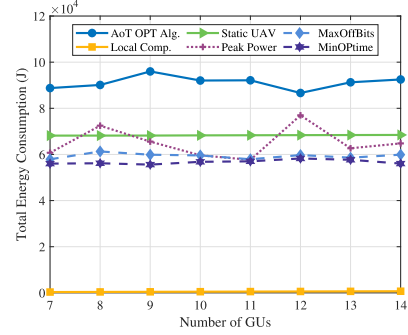


Fig. 8. Total Energy Consumption with varying $K \in [7, 14]$ when $L_k = 50$ Mb.

AoT because they prefer to improve offloading bits or time cost rather than reduce AoT.

C. Min-Max AoT Versus Task-Input Data Size L_k

Fig. 7 shows the effect of the task-input size L_k on the min-max AoT, where $K = 10$ GUs are randomly distributed. The min-max AoT of the proposed algorithm increases from 4.5×10^7 to 3.8×10^9 , when the size of the task-input data increases from 10 Mb to 100 Mb. We can also find that our proposed algorithm (i.e., AoT OPT Alg.) provides the smaller max-AoT for all these 10 GUs than the other benchmark schemes (including Local Comp., Peak Power and Static UAV) because of the joint optimization about UAV trajectories and resource allocations. Furthermore, the ‘Upper Bound’ based on Lemma 1 is consistent with the Local Comp. scheme in terms of the min-max AoT, because both of them only consider the computation capacity of GUs. Indeed, the ‘Lower Bound’ based on Lemma 1 is derived from a very ideal assumption, in which all the data transmission between GUs and UAVs will never result in the co-channel interference. Thus the gap between the ‘Lower Bound’ and the four available schemes is relatively huge compared with the ‘Upper Bound’. We can also find that the MaxOffBits scheme fluctuates more than other schemes. This is because it tries to offload as many bits as possible but ignores the bit’s completion time, which makes its iterations sometimes stop at the suboptimal solution in terms of AoT due to the same amount of offloading bits.

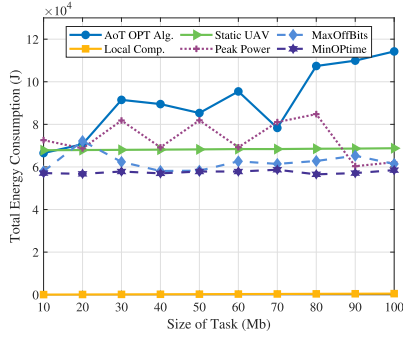


Fig. 9. Total Energy Consumption with varying $L_k \in [10, 100]$ Mb when $K = 10$ GUs.

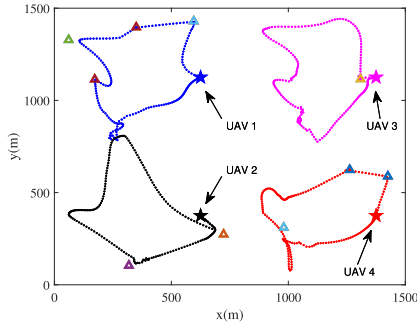


Fig. 10. UAV trajectories when $L_k = 100$ Mb and $E_{prop}^{\max} = 50,000$ J.

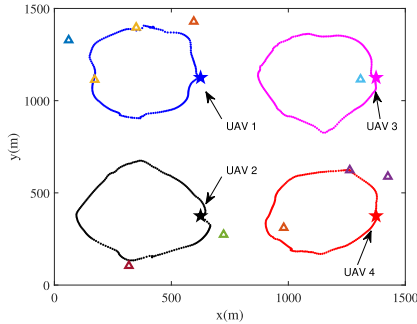


Fig. 11. UAV Trajectories with $L_k = 100$ Mb and $E_{prop}^{\max} = 16,000$ J.

D. Energy Consumption of Various Schemes

Besides the above temporal performance, we further evaluate the total energy consumption of different methods. The total energy consumption consists of the UAVs' propulsion energy, the UAVs' computation offloading energy, the GUs' communication energy and the GUs' local computation energy. Fig. 8 shows the total energy consumption when the number of GUs ranges from 7 to 14. Since the proposed method only focuses on the quality of experience from the perspective of users, the proposed method ignores other metrics and consumes the most amount of energy compared with other schemes. The total energy consumption with increasing task-input data size is presented in Fig. 9, where the energy consumption of the proposed algorithm increases from 70,000 J to 120,000 J with L_k increasing from 10 Mb to 100 Mb. Moreover, the local computation scheme always consumes the least energy, because the major

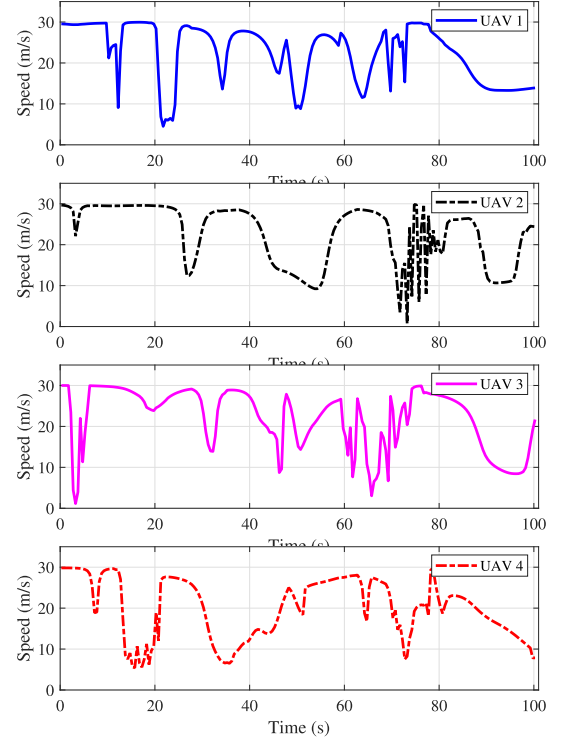


Fig. 12. UAV flying speed when $L_k = 100$ Mb and $E_{prop}^{\max} = 50,000$ J.

part of the total energy consumption is the propulsion energy of all the UAVs.

E. Case Study: Heavy-Weight Task $L_k = 100$ Mb in a Broad Area (1500×1500 M)

In order to validate the effect of our proposed algorithm, we plot the obtained UAV trajectories as a case study, which is shown in Fig. 10. Here the task-input size L_k and the UAV propulsion energy constraint $E_{prop}^{\max}$ are set as 100 Mb and 50,000 J. In Fig. 10, '★' and '△' represent the UAV startup coordinate and the GU location, respectively. The dense dots in Fig. 10 form four UAV trajectories obtained by the proposed algorithm. We can find that the UAVs tend to fly close to the GUs to improve the transmission rate and accelerate the task offloading. According to the obtained trajectories, all the UAVs will also take off from their own startup location and return to their startup location lastly. Fig. 12 presents the corresponding flying speed of 4 UAVs in this experiment with $L_k = 100$ Mb and $E_{prop}^{\max} = 50,000$ J. In most of the time, the UAV speed fluctuates in the range from 10 m/s to 30 m/s, which conforms to the constraint of the maximum velocity, i.e., $v_{\max} = 30$ m/s.

In addition, we also conduct the experiment with a lower UAV propulsion energy constraint $E_{prop}^{\max} = 16,000$ J, with which the trajectories are shown in Fig. 11. We can find that the trajectories in Fig. 11 approach much more to the initial anticlockwise circles than the trajectories in Fig. 10. This is because the lower UAV propulsion energy restrains that UAVs flying too fast or even close to GUs in the predetermined period T of the experiment. This phenomenon can also be revealed in Fig. 13, where these UAVs fly with a velocity between 10 and 20 m/s in the most of

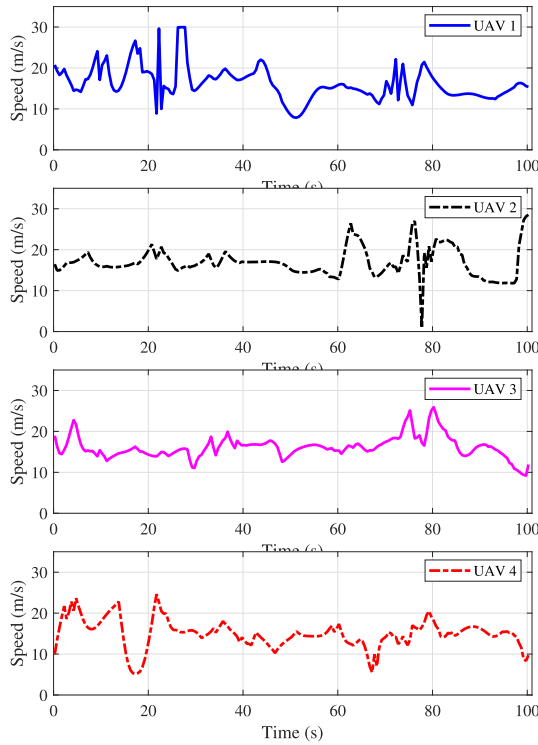


Fig. 13. UAV flying speed when $L_k = 100$ Mb and $E_{prop}^{\max} = 16,000$ J.

time. Thus the UAV velocities under the lower propulsion energy constraint $E_{prop}^{\max}$ are lower than that under the higher $E_{prop}^{\max}$.

VI. CONCLUSION

In this paper, we formulate a maximum AoT minimizing problem in the multiple UAV assisted MEC network. Here AoT is proposed to describe the user's temporal experience (in terms of task bit) more accurately than the common task latency, especially for the partial task offloading scenario. First, with the time discretization (i.e., the whole period is divided into several discrete time slots), the AoT metric can be approximately expressed via the amount of the bits processed at the specific slot. Meanwhile, the time discretization simplifies the mathematical formulation of the UAVs' trajectories and the co-channel interference among UAVs. Then we adopt the BCD method to optimize the UAV trajectory, the GU transmit power and the GU-MEC association iteratively. Moreover, the SCA technique is exploited in the sub-problems to transform the non-convex constraints into the convex ones. Thus we propose the overall joint optimization algorithm to minimize the maximum AoT of all the GUs (i.e., min-max AoT) based on the common convex optimization solver. Finally, the experimental results demonstrate that the proposed algorithm can effectively provide smaller AoT for the randomly distributed GUs compared with the benchmark schemes.

The proposed scheme is appropriate for offloading the continuous small tasks in the multi-UAV assisted MEC scenarios, where UAVs can provide the offloading services while flying.

However, there are still some unsolved issues, e.g., task dependence, security/privacy concerns, and energy efficiency. In the future, we will combine the user's energy consumption with the user's temporal experience in the multi-UAV scenario, where the task dependence (i.e., the offloading constraint based on DAG), the task-input data security and the user identity privacy are considered at the same time.

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